A Branch-and-Price Algorithm for Combined Location and Routing Problems Under Capacity Restrictions

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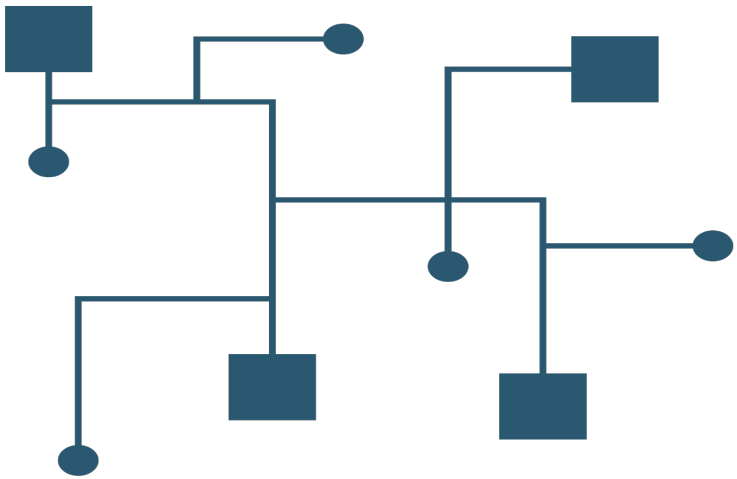


INDUSTRIAL AND SYSTEMS ENGINEERING

LEHIGH UNIVERSITY

October 29, 2020

Outline

1. Introduction
2. Problem Definition
3. Solution Methods

Introduction

Motivation: where to locate the facilities and how to allocate customers to the selected facilities with capacity constraints on both the facilities and the vehicles to minimize the total cost.

Literature

- Heuristics
  - Construction and Improvement
  - Clustering Algorithms
  - Metaheuristics
- Optimization-Based Approaches
  - Benders' Decomposition
  - Dynamic Programming
  - Polyhedral Approach
  - Matching-Based Approach
  - Column Generation

Problem Definition and Formulations

Objective: to determine the number and location of facilities and construct a set of vehicle routes from facilities to customers to minimize total cost.

Total cost: both the fixed and operating costs of facilities and vehicles.

Two Types of formulation:

- Vehicle Flow Formulation
- Set-Partitioning Formulation

Vehicle Flow Formulation[1]

Notations:

I : the set of customers,

J : the set of candidate facilities,

$V = I \cup J$,

Extra assumption: assume that each facility has its own set of vehicles,

H_j : the set of vehicles located at facility j, $\forall j \in J$,

H: the set of all vehicles,

[1] Z Akca, RT Berger, and TK Ralphs. "A branch-and-price algorithm for combined location and routing problems under capacity restrictions". In: *Operations research and cyber-infrastructure*. Springer, 2009, pp. 309–330.

Vehicle Flow Formulation-Cont

Parameters:

$$
\begin{aligned}
D_i &= \text{demand of customer } i, \forall i \in I, \\
C_j^F &= \text{capacity of facility } j, \forall j \in J, \\
C^V &= \text{capacity of a vehicle}, \\
F_j &= \text{fixed cost of opening facility } j, \forall j \in J, \\
O_{ik} &= \text{operating cost of traveling arc } (i,k)\ \forall i,k \in V.
\end{aligned}
$$

Decision Variables:

$$
\begin{aligned}
x_{ikh} &= \begin{cases} 1 & \text{if vehicle } h \text{ travels directly from location } i \text{ to location } k, \forall i \in V, k \in V, h \in H \\ 0 & \text{otherwise}, \end{cases} \\
y_{ih} &= \begin{cases} 1 & \text{if vehicle } h \text{ visits customer } i, \forall i \in I, h \in H \\ 0 & \text{otherwise}, \end{cases} \\
t_j &= \begin{cases} 1 & \text{if facility } j \text{ is selected to be open}, \forall j \in J \\ 0 & \text{otherwise}. \end{cases}
\end{aligned}
$$

Vehicle Flow Formulation-Cont

$$
\begin{aligned}
\text{Minimize} \quad & \sum_{j\in J} F_j t_j + \sum_{h\in H}\sum_{i\in V}\sum_{k\in V} O_{ik} x_{ikh} && && (1)\\
\text{s.t.} \quad & \sum_{h\in H}\sum_{k\in V} x_{ikh} = 1 && \forall i \in I, && (2)\\
& \sum_{k\in V} x_{ikh} - \sum_{k\in V} x_{kih} = 0 && \forall i \in V, h \in H, && (3)\\
& \sum_{k\in V} x_{kih} - y_{ih} = 0 && \forall i \in I, h \in H, && (4)\\
& y_{ih} - \sum_{k\in S}\sum_{l\in V\setminus S} x_{klh} \le 0 && \forall S \subseteq I, i \in S, h \in H, && (5)\\
& \sum_{i\in I} D_i y_{ih} - C^V \le 0 && \forall h \in H, && (6)\\
& \sum_{h\in H_j}\sum_{i\in I} D_i y_{ih} - C_j^F t_j \le 0 && \forall j \in J, && (7)\\
& x_{ikh} \in \{0,1\} && \forall i,k \in V, h \in H, && (8)\\
& y_{ih} \in \{0,1\} && \forall i \in I, h \in H, && (9)\\
& t_j \in \{0,1\} && \forall j \in J. && (10)
\end{aligned}
$$

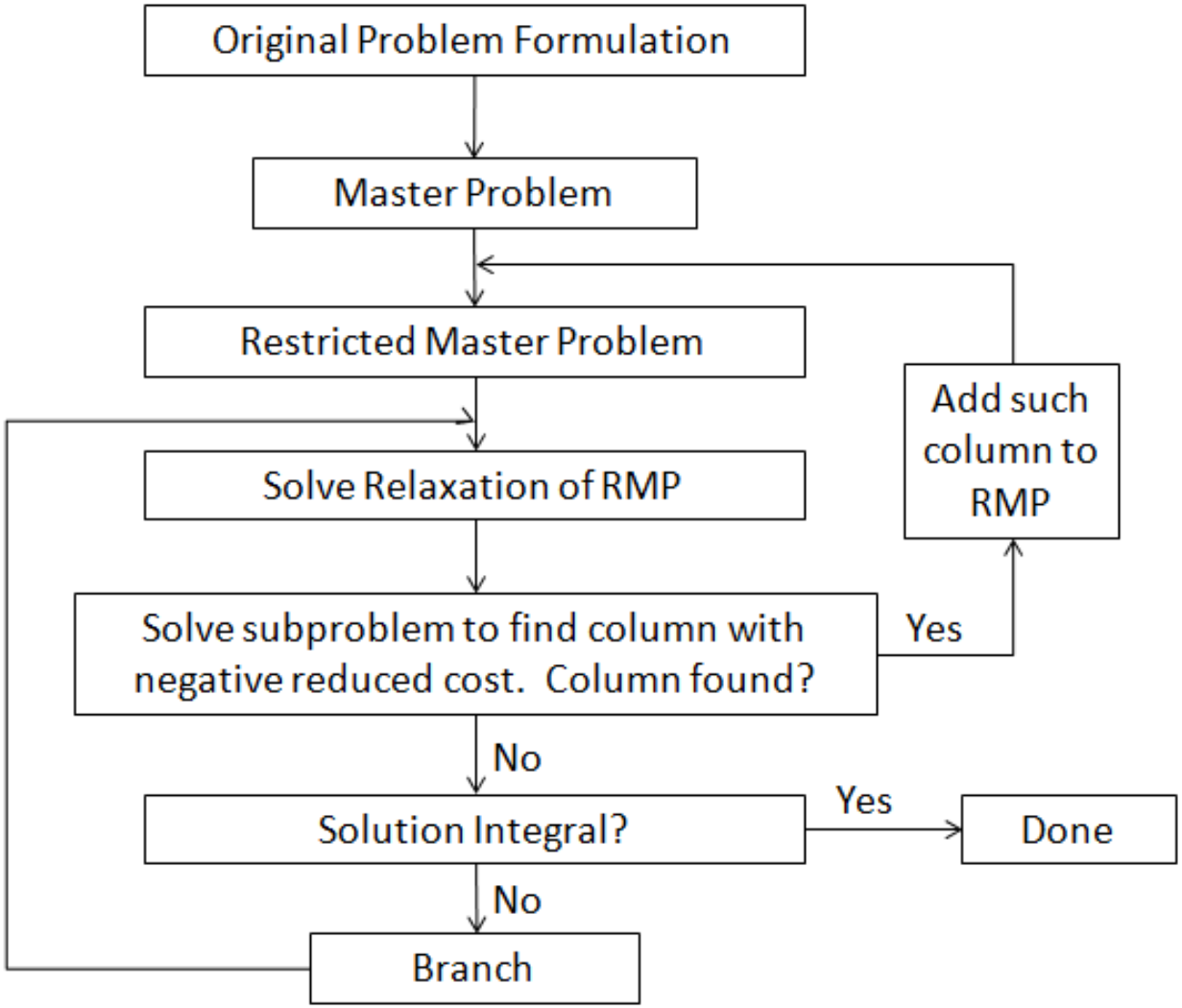
Branch and Price Framework

- The number of routes will be too large!
- To solve the LP relaxation of models with exponentially many columns, column generation algorithms can be used.
- By adding eligible columns dynamically to enter the basis, we generate only those columns that may improve the objective function.

- We used DipPy from COIN-OR[2] organization as the framework.
- DipPy itself reformulate our problem to the set-partitioning formulation.
  - The Restricted Master Problem (RMP) is the set-partitioning formulation with a subset of routes.
  - The subproblem is to find a route for a specific vehicle that can improve the objective of the RMP.

[2] https://www.coin-or.org/.

Branch and Price Framework[3]



[3] https://en.wikipedia.org/wiki/Branch_and_price.

Branch and Price Framework

$$\sum_{j\in J} F_j t_j + \sum_{h\in H}\sum_{i\in V}\sum_{k\in V} O_{ik} x_{ikh} \tag{1}$$

$$\sum_{h\in H}\sum_{k\in V} x_{ikh} = 1 \qquad \forall i \in I, \tag{2}$$

Linking

$$\sum_{k\in V} x_{ikh} - \sum_{k\in V} x_{kih} = 0 \qquad \forall i \in V, h \in H, \tag{3}$$

$$\sum_{k\in V} x_{kih} - y_{ih} = 0 \qquad \forall i \in I, h \in H, \tag{4}$$

Blocks

$$y_{ih} - \sum_{k\in S}\sum_{l\in V\setminus S} x_{klh} \leq 0 \qquad \forall S \subseteq I, i \in S, h \in H, \tag{5}$$

$$\sum_{i\in I} D_i y_{ih} - C^V \leq 0 \qquad \forall h \in H, \tag{6}$$

$$\sum_{h\in H_j}\sum_{i\in I} D_i y_{ih} - C_j^F t_j \leq 0 \qquad \forall j \in J, \tag{7}$$

Linking

$$x_{ikh} \in \{0,1\} \qquad \forall i,k \in V, h \in H, \tag{8}$$

$$y_{ih} \in \{0,1\} \qquad \forall i \in I, h \in H, \tag{9}$$

$$t_j \in \{0,1\} \qquad \forall j \in J. \tag{10}$$

Some Modifications

I ignored the subtour elimination constraints since I solved my subproblem with dynamic programming.

In addition, I added two other constraints:

- These constraints prevent assigning the vehicles that are specified for the specific facility to the route originating from other facilities:

$$X_{jkh} = 0 \quad j \in J, k \in I, h \in (H \setminus H_j),$$

- and these constraints prevent self cycling:

$$X_{iih} = 0 \quad i \in V, h \in H.$$

Solve Subproblem-Exact

Our subproblem: the Elementary Shortest Path Problem with Resource Constraints (ESPPRC).

We used Feillet's Label Setting algorithm[4] for solving ESPPRC.

Some basics of Feillet's Algorithm:

A node is unreachable:

- If they have already been visited in the path, or
- If they cannot be served by the vehicle without violating a resource constraint.

[4] Dominique Feillet et al. "An exact algorithm for the elementary shortest path problem with resource constraints: Application to some vehicle routing problems". In: *Networks: An International Journal* 44.3 (2004), pp. 216–229.

Solve Subproblem-Exact-Cont.

For each node we have list of labels like $\{L_1^i, L_2^i, \ldots, L_\mu^i\}$ in which there are information about one path of node i.

$L_1^i =< R_C^i, R_V^i, R_n^i, V_1^i, \ldots, V_I^i >$

$R_C^i =$ Total cost so far,

$R_V^i =$ Total demands so far,

$R_n^i =$ Total number of unreachable nodes,

V_k^i: status of going from i to k,

$$V_k^i = \begin{cases} -1, & \text{if node } k \text{ is unreachable due to capacity constraint,} \\ 0, & \text{if node } k \text{ is reachable and is not in the path so far,} \\ s > 0, & \text{if node } k \text{ is the s'th node in the path.} \end{cases}$$

Solve Subproblem-Exact-Cont.

- Initialization Step:
  - Label list is created for each node to keep the generated labels.
  - All label lists (except the list for the source node) are initially empty.
  - The source has one label, which is a zero vector.

Solve Subproblem-Exact-Cont.

- Extension step
  - For each node, all unprocessed labels are extended to all possible successors. The label l_r^i for node i can be extended to node k if
    - node k is reachable, i.e., $V_k^i = 0$,
    - the path is feasible, i.e., $R_{kv} + D_g \leq C^k$.
  - If it is feasible to extend l_r^i to node k, a new label is created for node k by updating the resource consumptions as follows:
    - $R_{ck} = R_c^i + C_{ik}$
    - $R_{vk} = R_v^i + D_k$
    - We update the status of nodes on the labels as follows:

$$V_g^k = \begin{cases} V_g^i, & \text{if } V_g^i \neq 0, \\ 0, & \text{if } V_g^i = 0 \text{ and } R_v^k + D_g > C^k, \\ N, & \text{if } g = k \text{ and } N \text{ is the \# of nodes in path}, \\ 0, & \text{o/w.} \end{cases}$$

Solve Subproblem-Exact-Cont.

- Elimination step: After extension step we apply dominance properties rule to the newly generated labels. Label 1 dominates label 2, if:
  - $(R_v^k)^1 \leq (R_v^k)^2$,
  - $(R_C^k)^1 \leq (R_C^k)^2$,
  - if node g is unreachable in 1, it is unreachable in 2.

Solve Subproblem-Heuristics.

- Heuristic_CS: We set the limit to the size of the customers considered in each pricing problem with *minimum spanning tree* algorithm.
- Heuristic_LL: We set the limit on the size of the labels in each node.

Computational Experiments

	B&P	*Heur_CS5*		*Heur_CS10*		*Heur_LL5*		*Heur_LL10*	
ID	*Time (sec)*	*Time (sec)*	*Opt. Gap*	*Time (sec)*	*Opt. Gap*	*Time (sec)*	*Opt. Gap*	*Time (sec)*	*Opt. Gap*
r15x6_a_1	**857**	**1581**	**0.00**	**933**	**0.00**	**1601**	**0.00**	**778**	**0.00**
r15x6_a_2	**732**	530	0.02	**558**	**0.00**	630	0.02	**541**	**0.00**
r15x6_a_3	**1167**	**759**	**0.00**	**885**	**0.00**	**736**	**0.00**	**935**	**0.00**
r15x6_a_4	**1042**	**731**	**0.00**	**611**	**0.00**	**627**	**0.00**	**600**	**0.00**
r15x6_a_5	**691**	**558**	**0.00**	**684**	**0.00**	**563**	**0.00**	**673**	**0.00**
r15x6_a_6	**14891**	**665**	**0.00**	**2250**	**0.00**	**658**	**0.00**	**2231**	**0.00**
r15x6_a_7	**84933**	4402	0.06	**17494**	**0.00**	5870	0.07	**17489**	**0.00**
r15x6_a_8	**7099**	11317	0.08	**1508**	**0.00**	1908	0.10	**1483**	**0.00**
r15x6_b_3	**210084**	**1056**	**0.00**	**17508**	**0.00**	**918**	**0.00**	**17484**	**0.00**
r20x8_a_2	**9121**	7085	0.02	8195	0.01	5741	0.02	7275	0.01
r20x8_a_3	**8433**	7029	0.08	**7254**	**0.00**	7263	0.08	**6216**	**0.00**

$$C^V = (1-\alpha)\max_{i\in I}(D_i) + \alpha(\sum_{i\in I} D_i)$$

$$\alpha = \{0, 0.25, 0.5, 0.75, 1\}$$

Some Observations

- Heuristic method CS5:
  - Average optimality gap of the solution is 5%,
  - About 50.6% reduction in execution time.
- Heuristic method CS10:
  - Average optimality gap of the solution is 0.1%,
  - About 40% reduction in execution time.
- Heuristic method LL5:
  - Average optimality gap of the solution is 3.2%,
  - About 32.4% reduction in execution time.
- Heuristic method LL10:
  - Average optimality gap of the solution is 0.02%,
  - About 9.1% reduction in execution time.

Questions?

Thank you!